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-- BEGINNER EXERCISES
-- Topics: SELECT, WHERE, ORDER BY, LIMIT, DISTINCT, LIKE, NULL
-- =====

-- Exercise 1
-- Get the first name, last name, and job title of all employees.
select first_name, last_name, job_title from employees
```

first_name varchar	last_name varchar	job_title varchar
Sam	CEO	CEO
Alice	Smith	VP Engineering
David	Brown	VP Sales
Grace	Wilson	HR Director
Iris	Taylor	Marketing Director
Karen	Thomas	CFO
Bob	Jones	Senior Engineer
Eva	Davis	Sales Manager
Leo	Jackson	Finance Manager
Carol	White	Engineer
.	.	.
.	.	.
.	.	.
Henry	Moore	HR Specialist
Rachel	Scott	HR Specialist
Steve	King	HR Analyst
Jack	Anderson	Marketing Analyst
Tina	Wright	Marketing Analyst
Uma	Lopez	Senior Marketing Analyst
Victor	Hill	Financial Analyst
Wendy	Green	Financial Analyst
Xander	Baker	Engineer
Yara	Nelson	Junior Engineer

26 rows (20 shown)

3 columns

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-- Exercise 2
-- Get all employees who work in department_id = 1.
select * from employees where department_id = 1
```

id int32	first_name varchar	last_name varchar	email varchar	department_id int32	manager_id int32	salary decimal(10,2)	hire_date date	job_title varchar
16	Sam	CEO	sam@company.com	1	NULL	200000.00	2010-01-01	CEO
1	Alice	Smith	alice@company.com	1	16	120000.00	2018-03-15	VP Engineering
2	Bob	Jones	bob@company.com	1	1	90000.00	2019-06-01	Senior Engineer
3	Carol	White	carol@company.com	1	2	85000.00	2020-01-10	Engineer
13	Mia	Harris	mia@company.com	1	2	88000.00	2020-03-25	Engineer
15	Olivia	Garcia	olivia@company.com	1	2	92000.00	2018-12-01	Senior Engineer
25	Xander	Baker	xander@company.com	1	2	82000.00	2021-09-05	Engineer
26	Yara	Nelson	yara@company.com	1	2	79000.00	2023-03-14	Junior Engineer

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-- Exercise 3
-- Get all employees with a salary greater than 90,000.
-- Show their full name and salary, sorted by salary highest to lowest.
select first_name, last_name, salary from employees where salary > 90000 order by salary desc
```

first_name varchar	last_name varchar	salary decimal(10,2)
Sam	CEO	200000.00
Alice	Smith	120000.00
Karen	Thomas	115000.00
David	Brown	110000.00
Iris	Taylor	105000.00
Grace	Wilson	95000.00
Olivia	Garcia	92000.00

```
-- Exercise 4
-- Get the 5 most recently hired employees.
-- Show their name and hire_date.
select first_name, last_name, hire_date from employees order by hire_date desc limit 5
```

first_name varchar	last_name varchar	hire_date date
Quinn	Adams	2023-05-22
Yara	Nelson	2023-03-14
Jack	Anderson	2023-01-03
Nathan	Martin	2022-09-12
Wendy	Green	2022-07-11

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-- Exercise 5
-- Get all unique job titles in the company.
select DISTINCT job_title from employees
```

job_title varchar
Engineer
Financial Analyst
VP Sales
Finance Manager
HR Specialist
VP Engineering
Marketing Director
Marketing Analyst
CEO
Junior Engineer
Senior Marketing Analyst
Sales Manager
HR Director
Senior Engineer
HR Analyst
CFO
Sales Rep

17 rows

```
-- Exercise 6
-- Find all employees whose last name starts with the letter 'M'.
select first_name from employees where last_name LIKE 'M%'
```

first_name
varchar

Frank
Nathan
Henry

```
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-- Exercise 7
-- Find all employees who do NOT have a manager (manager_id is NULL).
select * from employees where manager_id is NULL
```

id int32	first_name varchar	last_name varchar	email varchar	department_id int32	manager_id int32	salary decimal(10,2)	hire_date date	job_title varchar
16	Sam	CEO	sam@company.com	1	NULL	200000.00	2010-01-01	CEO

```
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-- Exercise 8
-- Get all projects that are currently 'active'.
-- Show the project name, budget, and start date.
-- select name, budget, start_date from projects where status = 'active'
Error 'NoneType' object has no attribute 'show'
```

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-- commented out because status of projects is not available in the database.
```

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-- Exercise 9
-- Get all sales where the amount is between 10,000 and 25,000.
-- Sort by amount ascending.
select amount, product from sales where amount between 10000 and 25000 order by amount asc
```

amount decimal(10,2)	product varchar
10000.00	Starter Plan
11000.00	Starter Plan
11000.00	Pro Plan
12000.00	Pro Plan
12000.00	Pro Plan
13000.00	Pro Plan
13500.00	Pro Plan
14000.00	Pro Plan
14000.00	Pro Plan
15000.00	Pro Plan
.	.
.	.
.	.
18000.00	Pro Plan
18000.00	Pro Plan
19000.00	Pro Plan
19000.00	Pro Plan
20000.00	Enterprise
21000.00	Enterprise
22000.00	Enterprise
22000.00	Enterprise
23000.00	Enterprise
24000.00	Enterprise

24 rows (20 shown) 2 columns

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-- Exercise 10
-- Count how many employees are in the company total.
select count(*) as employee_count from employees
```

employee_count int64
26

```
-- Exercise 11
-- Get all employees hired in the year 2020.
-- Show their full name and hire_date.
select first_name, last_name, hire_date from employees where hire_date between '2020-01-01' and '2020-12-31'
```

first_name varchar	last_name varchar	hire_date date
Leo	Jackson	2020-10-07
Carol	White	2020-01-10
Mia	Harris	2020-03-25
Tina	Wright	2020-06-15

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-- Exercise 12
-- Get all sales made in the 'East' region.
-- Show the sale amount, product, and sale_date.
select amount, product, sale_date from sales where region = 'East'
```

amount decimal(10,2)	product varchar	sale_date date
15000.00	Pro Plan	2023-01-05
11000.00	Starter Plan	2023-01-20
22000.00	Enterprise	2023-02-08
16000.00	Pro Plan	2023-02-22
27000.00	Enterprise	2023-03-10
19000.00	Pro Plan	2023-04-03
8500.00	Starter Plan	2023-04-15
14000.00	Pro Plan	2023-04-28
7500.00	Starter Plan	2023-05-30
24000.00	Enterprise	2023-06-11
20000.00	Enterprise	2023-06-22
16000.00	Pro Plan	2023-07-04
28000.00	Enterprise	2023-08-08
23000.00	Enterprise	2023-08-21
19000.00	Pro Plan	2023-09-18
10000.00	Starter Plan	2023-10-09
26000.00	Enterprise	2023-10-23
12000.00	Pro Plan	2023-11-28
32000.00	Enterprise	2023-12-05
9000.00	Starter Plan	2023-12-18

```
-- Exercise 13
-- Show all employees whose salary is NOT between 70,000 and 90,000.
select first_name from employees where salary not between 70000 and 90000
```

first_name varchar

Sam
Alice
David
Grace
Iris
Karen
Olivia
Nathan
Quinn
Henry
Rachel
Steve

12 rows

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-- Exercise 14
-- Get all projects that have an end_date (i.e. end_date is not NULL).
-- Show project name and end_date, sorted by end_date ascending.
-- select name, end_date from projects where end_date is not NULL order by end_date asc
Error 'NoneType' object has no attribute 'show'
```

```
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-- commented out because end_date of projects is not available in the database.
```

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-- Exercise 15
-- Find all employees whose first name contains the letter 'a' (case-insensitive).
select first_name from employees where lower(first_name) LIKE '%a%'
```

first_name varchar
Sam
Alice
David
Grace
Karen
Eva
Carol
Mia
Olivia
Frank
Nathan
Paula
Rachel
Jack
Tina
Uma
Xander
Yara

18 rows

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-- Exercise 16
-- Show the total salary paid across the entire company.
select sum(salary) as total_salary from employees
```

total_salary decimal(38,2)
2262000.00

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-- Exercise 17
-- Get all employees sorted by department_id ascending,
-- then by salary descending within each department.
select * from employees order by department_id, salary desc

```

id int32	first_name varchar	last_name varchar	email varchar	department_id int32	manager_id int32	salary decimal(10,2)	hire_date date	job_title varchar
16	Sam	CEO	sam@company.com	1	NULL	200000.00	2010-01-01	CEO
1	Alice	Smith	alice@company.com	1	16	120000.00	2018-03-15	VP Engineering
15	Olivia	Garcia	olivia@company.com	1	2	92000.00	2018-12-01	Senior Engineer
2	Bob	Jones	bob@company.com	1	1	90000.00	2019-06-01	Senior Engineer
13	Mia	Harris	mia@company.com	1	2	88000.00	2020-03-25	Engineer
3	Carol	White	carol@company.com	1	2	85000.00	2020-01-10	Engineer
25	Xander	Baker	xander@company.com	1	2	82000.00	2021-09-05	Engineer
26	Yara	Nelson	yara@company.com	1	2	79000.00	2023-03-14	Junior Engineer
4	David	Brown	david@company.com	2	16	110000.00	2017-11-20	VP Sales
5	Eva	Davis	eva@company.com	2	4	75000.00	2021-04-05	Sales Manager
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.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
8	Henry	Moore	henry@company.com	3	7	65000.00	2022-02-14	HR Specialist
19	Rachel	Scott	rachel@company.com	3	7	63000.00	2021-11-30	HR Specialist
9	Iris	Taylor	iris@company.com	4	16	105000.00	2019-08-22	Marketing Director
22	Uma	Lopez	uma@company.com	4	9	78000.00	2018-08-20	Senior Marketing Analyst
21	Tina	Wright	tina@company.com	4	9	74000.00	2020-06-15	Marketing Analyst
10	Jack	Anderson	jack@company.com	4	9	70000.00	2023-01-03	Marketing Analyst
11	Karen	Thomas	karen@company.com	5	16	115000.00	2015-05-18	CFO
12	Leo	Jackson	leo@company.com	5	11	80000.00	2020-10-07	Finance Manager
23	Victor	Hill	victor@company.com	5	12	76000.00	2021-02-28	Financial Analyst
24	Wendy	Green	wendy@company.com	5	12	73000.00	2022-07-11	Financial Analyst

26 rows (20 shown)

9 columns

```

-- Exercise 18
-- Find all sales for the product 'Enterprise'.
-- Show employee_id, amount, and sale_date.
select employee_id, amount, sale_date from sales where product = 'Enterprise'

```

employee_id int32	amount decimal(10,2)	sale_date date
5	22000.00	2023-02-08
14	27000.00	2023-03-10
17	21000.00	2023-03-18
6	31000.00	2023-05-07
14	24000.00	2023-06-11
18	20000.00	2023-06-22
14	28000.00	2023-08-08
5	23000.00	2023-08-21
5	26000.00	2023-10-23
17	22000.00	2023-11-19
14	32000.00	2023-12-05

11 rows

3 columns

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-- Exercise 19
-- Show the minimum, maximum, and average salary in the company.
-- Label the columns as min_salary, max_salary, avg_salary.
select min(salary) as min_salary, max(salary) as max_salary, avg(salary) as avg_salary from employees

```

min_salary decimal(10,2)	max_salary decimal(10,2)	avg_salary double
63000.00	200000.00	87000.0

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-- Exercise 20

-- Get all departments located in 'New York'.

select * from departments where location = 'New York'

id int32	name varchar	location varchar
1	Engineering	New York
3	HR	New York